

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 4: Vector Spaces

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October 7, 2026

1 Keywords

- Fields, vector spaces, metrics, Lipschitz functions, norms, spectral norm, regularisation

2 Main Points

2.1 Scalars (Fields)

- A vector space is built over a set of *scalars*, formally a *field*: numbers we can add, subtract, multiply and divide (except by 0) with the usual rules
- The rationals, the reals and the complex numbers are fields. The integers are not, since 2 has no integer inverse
- We will almost always use the reals

2.2 Vector Spaces

- A set of objects $\mathcal{V}$, with addition and multiplication by a scalar, is a *vector space* if
 1. $a v \in \mathcal{V}$ and $v + w \in \mathcal{V}$ for all $v, w \in \mathcal{V}$ (closure)
 2. $v + w = w + v$ (commutativity of addition)
 3. $(u + v) + w = u + (v + w)$ (associativity of addition)
 4. $v + \mathbf{0} = v$ (additive identity $\mathbf{0}$)
 5. $v + (-v) = \mathbf{0}$ (additive inverse $-v$)
 6. $1 v = v$ (multiplicative identity)
 7. $a(b v) = (a b) v$ (associativity of scalar multiplication)
 8. $a(v + w) = a v + a w$ and $(a + b) v = a v + b v$ (distributivity)
- Most of these hold automatically if addition and scalar multiplication behave like ordinary arithmetic. The one to watch is closure: adding two members, or scaling one, must not take you outside the set
- Column vectors in $\mathbb{R}^n$, matrices of a fixed shape, infinite sequences and functions on an interval all form vector spaces. Anything satisfying the axioms is a "vector"
- Bounded vectors (say $\|v\| \leq 1$) do *not* form a vector space: $2v$ may fall outside the set

2.3 Distances (Metrics)

- A *distance* or *metric* satisfies 4 conditions
 1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ (non-negativity)
 2. $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$ (identity of indiscernibles)
 3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ (symmetry)
 4. $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$ (triangular inequality)
- You can define different distances for the same set of objects (e.g. Euclidean and Manhattan distance)
- Often we use *pseudo-metrics* that break one or other of the conditions

2.4 Mappings and Lipschitz Functions

- A function $f : \mathcal{V} \rightarrow \mathcal{W}$ maps one vector space to another (possibly the same one). Mappings compose: for matrices, $\mathbf{A} \mathbf{B} \mathbf{x} = \mathbf{A} (\mathbf{B} \mathbf{x})$ means "apply $\mathbf{B}$, then $\mathbf{A}$ "
- A useful measure of how much a mapping can amplify changes is the *Lipschitz condition*: f is K -Lipschitz if, for all x and y ,

$$d(f(x), f(y)) \leq K d(x, y)$$

for some fixed $K < \infty$

- Lipschitz functions are continuous (they have no jumps), but not every continuous function is Lipschitz: $\sqrt{x}$ on $[0, 1]$ is continuous but infinitely steep at 0
 - For a differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$, the smallest K is the largest value of $|f'(x)|$
- If $K < 1$ the mapping is *contractive*. The *Banach fixed-point theorem* says a contractive mapping (of $\mathbb{R}^n$, or any complete metric space, into itself) has exactly one fixed point $x^* = f(x^*)$, and iterating $x_{t+1} = f(x_t)$ from any start converges to it. Each step shrinks the distance to x^* by at least a factor K : $d(x_{t+1}, x^*) = d(f(x_t), f(x^*)) \leq K d(x_t, x^*)$. This is a standard way to prove that an iterative algorithm converges

2.5 Norms

- Norms provide a measure of the size of a vector. They satisfy three conditions
 1. $\|\mathbf{v}\| > 0$ if $\mathbf{v} \neq \mathbf{0}$ and $\|\mathbf{0}\| = 0$ (non-negativity)
 2. $\|a \mathbf{v}\| = |a| \|\mathbf{v}\|$ ("linearity")
 3. $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangular inequality)
- If not all of these conditions hold we have a *pseudo-norm*
- **Vector norms**
 - Common examples
 1. Euclidean or 2-norm: $\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$
 2. p -norm ($p \geq 1$): $\|\mathbf{v}\|_p = (\sum_{i=1}^n |v_i|^p)^{1/p}$
 3. 1-norm: $\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$
 4. ∞ -norm or max-norm: $\|\mathbf{v}\|_\infty = \max_i |v_i|$
 - The 1-, 2- and ∞ -norms are all p -norms
 - The "0-norm", the number of non-zero components, is only a pseudo-norm, as $\|2 \mathbf{v}\|_0 = \|\mathbf{v}\|_0 \neq 2 \|\mathbf{v}\|_0$

- **Norms for regularisation.** When many weight vectors $\mathbf{w}$ fit the training data equally well we usually prefer a "small" one, so we minimise $\text{loss}(\mathbf{w}) + \eta \|\mathbf{w}\|$. This is equivalent to minimising the loss subject to $\|\mathbf{w}\| \leq c$: confining $\mathbf{w}$ to a *ball* of the chosen norm. Each norm means something different by small

- $\|\mathbf{w}\|_2$ small: no component is large; the weight is spread out
- $\|\mathbf{w}\|_1$ small: a few large components are tolerated if the rest are tiny. The 1-norm ball is a diamond with corners on the axes, and the optimum often lands on a corner where most components are exactly zero. This is why L_1 regularisation (the *lasso*) picks out a few features
- $\|\mathbf{w}\|_\infty$ small: only the largest component is controlled
- Choosing a norm is choosing what you mean by "small"

- **Matrix norms**

- Examples

1. The Frobenius norm is $\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$
2. The *spectral norm* is the largest factor by which $\mathbf{A}$ can stretch a vector

$$\|\mathbf{A}\|_2 = \max_{\mathbf{v} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{v}\|_2}{\|\mathbf{v}\|_2}$$

For a symmetric matrix this is its largest $|\lambda|$. For a general matrix it is the largest singular value (see the SVD lecture), which can be bigger than any $|\lambda|$

3. There are many others: the 1-norm, max-norm, nuclear norm, ...
- The determinant is not a norm: it can be negative and does not scale correctly
 - Many (but not all) matrix norms satisfy

$$\|\mathbf{A}\mathbf{B}\| \leq \|\mathbf{A}\| \times \|\mathbf{B}\|$$

which lets us bound the norm of a product quickly

- A matrix norm $\|\cdot\|_a$ and a vector norm $\|\cdot\|_b$ are *compatible* if

$$\|\mathbf{M}\mathbf{v}\|_b \leq \|\mathbf{M}\|_a \times \|\mathbf{v}\|_b$$

The spectral and Euclidean norms are compatible; this is just the definition of the spectral norm. The Frobenius and Euclidean norms are compatible too, since $\|\mathbf{M}\|_2 \leq \|\mathbf{M}\|_F$

- The sensitivity of a mapping is measured by its *condition number* $\|\mathbf{M}\| \|\mathbf{M}^{-1}\|$: the ratio of its largest to smallest stretch (see the eigensystems notes)

2.6 The Lipschitz Constant of a Network

- A deep network applies layers in turn. Ignoring biases, a linear layer is $\mathbf{x}_n = \mathbf{L}_n \mathbf{x}_{n-1}$, interleaved with non-linearities such as ReLUs and pooling
- Lipschitz constants multiply under composition: if g is K_g -Lipschitz and f is K_f -Lipschitz then $\|f(g(\mathbf{x})) - f(g(\mathbf{y}))\| \leq K_f \|g(\mathbf{x}) - g(\mathbf{y})\| \leq K_f K_g \|\mathbf{x} - \mathbf{y}\|$
- By compatibility, a linear layer has $K = \|\mathbf{L}_n\|_2$. ReLUs and max-pooling never stretch distances (they can shrink them), so $K = 1$. Hence

$$K_{\text{net}} \leq \prod_n \|\mathbf{L}_n\|_2$$

This bound can be very loose (see Exercise 5)

- **Why care?** $\|f(x + \delta) - f(x)\| \leq K \|\delta\|$ bounds how much the output can move when the input is nudged
 - *Adversarial examples* are tiny δ that change the classification; a network with small K is harder to fool
 - $K \gg 1$ lets signals (and gradients) explode; $K \ll 1$ lets them vanish
 - Wasserstein GANs need a critic with $K \leq 1$. One way to enforce this is *spectral normalisation*: divide each $\mathbf{L}_n$ by $\|\mathbf{L}_n\|_2$
 - Some bounds on how well a network generalises depend on K
 - But make K too small and the network is too smooth to fit the data

2.7 Function Norms

- The most common function norms, for f defined on a region $\mathcal{R}$, are

1. The L_2 -norm

$$\|f\|_{L_2} = \sqrt{\int_{\mathbf{x} \in \mathcal{R}} f^2(\mathbf{x}) \, d\mathbf{x}}$$

2. The L_1 -norm

$$\|f\|_{L_1} = \int_{\mathbf{x} \in \mathcal{R}} |f(\mathbf{x})| \, d\mathbf{x}$$

3. The ∞ or max-norm

$$\|f\|_{\infty} = \max_{\mathbf{x} \in \mathcal{R}} |f(\mathbf{x})|$$

- Function norms also define vector spaces: the L_2 space is the set of functions with $\|f\|_{L_2} < \infty$, and similarly for L_1
- In these spaces two functions that differ only on a set of zero size (e.g. at a few isolated points) count as the same function, so that $\|f\| = 0$ only for the zero function

2.8 Norms and Metrics

- Every norm induces a distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$, so every normed space is also a metric space (Exercise 6). For example the Frobenius distance between matrices is $d_F(\mathbf{A}, \mathbf{B}) = \|\mathbf{A} - \mathbf{B}\|_F$
- Next lecture adds an *inner product*, which gives us angles as well as lengths

3 Exercises

1. Which of these are vector spaces (with the usual addition and scalar multiplication)? Give a reason
 - (a) Vectors in $\mathbb{R}^2$ with $x_1 \geq 0$
 - (b) Polynomials of degree at most 3
 - (c) Vectors in $\mathbb{R}^2$ with $x_1 + x_2 = 0$
 - (d) Vectors in $\mathbb{R}^2$ with $x_1 + x_2 = 1$
2. For $\mathbf{v} = (3, -4, 0, 1)^\top$ compute $\|\mathbf{v}\|_1$, $\|\mathbf{v}\|_2$, $\|\mathbf{v}\|_{\infty}$ and $\|\mathbf{v}\|_0$. Check that $\|\mathbf{v}\|_{\infty} \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1$

3. Find a Lipschitz constant K , or show none exists, for
 - (a) $f(x) = 3x + 1$ on $\mathbb{R}$
 - (b) $f(x) = \sin(x)$ on $\mathbb{R}$
 - (c) $f(x) = x^2$ on $\mathbb{R}$, and on $[-1, 1]$
 - (d) $\text{ReLU}(x) = \max(0, x)$ on $\mathbb{R}$
4. Show that $f(x) = x/2 + 1$ is contractive, find its fixed point, and iterate $x_{t+1} = f(x_t)$ from $x_0 = 0$. How fast does the error shrink?
5. Let $\mathbf{L}_1 = \begin{pmatrix} 3 & 0 \\ 0 & 0.5 \end{pmatrix}$ and $\mathbf{L}_2 = \begin{pmatrix} 0.5 & 0 \\ 0 & 3 \end{pmatrix}$
 - (a) Find $\|\mathbf{L}_1\|_2$ and $\|\mathbf{L}_1\|_F$
 - (b) Compare $\|\mathbf{L}_2 \mathbf{L}_1\|_2$ with the bound $\|\mathbf{L}_2\|_2 \|\mathbf{L}_1\|_2$. What does this say about the bound on K_{net} ?
6. Show that $d(x, y) = \|x - y\|$ satisfies the four conditions for a metric

4 Experiments

4.1 Norms

- In python, compare the spectral norm with the largest stretch of many random unit vectors, and with the largest eigenvalue

```
import numpy as np
```

```
v = np.array([3, -4, 0, 1.0])
print([np.linalg.norm(v, p) for p in (1, 2, np.inf)]) # 1-, 2- and max-norms
```

```
rng = np.random.default_rng(0)
A = rng.normal(size=(3, 3)) # not symmetric
u = rng.normal(size=(3, 100000))
u /= np.linalg.norm(u, axis=0) # 100000 random unit vectors
print(np.linalg.norm(A @ u, axis=0).max()) # largest stretch found
print(np.linalg.norm(A, 2)) # the spectral norm
print(np.abs(np.linalg.eigvals(A)).max()) # largest |eigenvalue|: smaller
print(np.linalg.norm(A, 'fro')) # Frobenius norm: larger
```

- The largest stretch found is (just below) the spectral norm, about 1.82. The largest $|\lambda|$ is only about 1.54: for a non-symmetric matrix the eigenvalues do not tell you how much it can stretch a vector

5 Answers

1. Vector spaces?
 - (a) No: $(1, 0)^\top$ is in the set but $-1 \times (1, 0)^\top$ is not (closure fails)
 - (b) Yes: sums and multiples of polynomials of degree at most 3 are polynomials of degree at most 3, and the zero polynomial is in the set
 - (c) Yes: if $x_1 + x_2 = 0$ and $y_1 + y_2 = 0$ then $(x_1 + y_1) + (x_2 + y_2) = 0$ and $a x_1 + a x_2 = 0$. It is a line through the origin
 - (d) No: it does not contain $\mathbf{0}$, and adding two members gives $x_1 + x_2 = 2$

2. $\|v\|_1 = 3 + 4 + 0 + 1 = 8$, $\|v\|_2 = \sqrt{9 + 16 + 0 + 1} = \sqrt{26} \approx 5.10$, $\|v\|_\infty = 4$ and $\|v\|_0 = 3$.
Indeed $4 \leq 5.10 \leq 8$

3. Lipschitz constants

(a) $|f(x) - f(y)| = 3|x - y|$, so $K = 3$

(b) $|f'(x)| = |\cos(x)| \leq 1$, so $K = 1$ (by the mean value theorem $|\sin x - \sin y| = |\cos \xi| |x - y|$ for some ξ)

(c) $|x^2 - y^2| = |x + y| |x - y|$ and $|x + y|$ is unbounded on $\mathbb{R}$, so no K works. On $[-1, 1]$, $|x + y| \leq 2$ so $K = 2$

(d) $K = 1$: if $x, y \geq 0$ the difference is unchanged; if both are negative it becomes 0; if $x \geq 0 > y$ then $|\text{ReLU}(x) - \text{ReLU}(y)| = x < x - y$. ReLU never stretches but can shrink distances

4. $|f(x) - f(y)| = \frac{1}{2}|x - y|$, so $K = \frac{1}{2} < 1$. The fixed point solves $x = x/2 + 1$, i.e. $x^* = 2$. From $x_0 = 0$: 1, 1.5, 1.75, 1.875, ... The error $x^* - x_t = 2^{1-t}$ halves at every step, exactly the factor K

5. Spectral norms of layers

(a) A diagonal matrix stretches the i -th axis by $|L_{ii}|$, so $\|\mathbf{L}_1\|_2 = 3$, while $\|\mathbf{L}_1\|_F = \sqrt{9 + 0.25} \approx 3.04$

(b) $\mathbf{L}_2 \mathbf{L}_1 = 1.5 \mathbf{I}$, so $\|\mathbf{L}_2 \mathbf{L}_1\|_2 = 1.5$, but the bound gives $3 \times 3 = 9$. Each layer stretches a different direction, and the second undoes the first. The product bound on K_{net} is safe but can be very pessimistic

6. Non-negativity and $d = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$ follow from norm condition 1. Symmetry: $\|\mathbf{x} - \mathbf{y}\| = \|(-1)(\mathbf{y} - \mathbf{x})\| = |-1| \|\mathbf{y} - \mathbf{x}\|$. Triangular inequality: $\|\mathbf{x} - \mathbf{y}\| = \|(\mathbf{x} - \mathbf{z}) + (\mathbf{z} - \mathbf{y})\| \leq \|\mathbf{x} - \mathbf{z}\| + \|\mathbf{z} - \mathbf{y}\|$